Differential Equations for Cyclical Systems — An Exploration of T. Tahara et al.'s Modified LV Model

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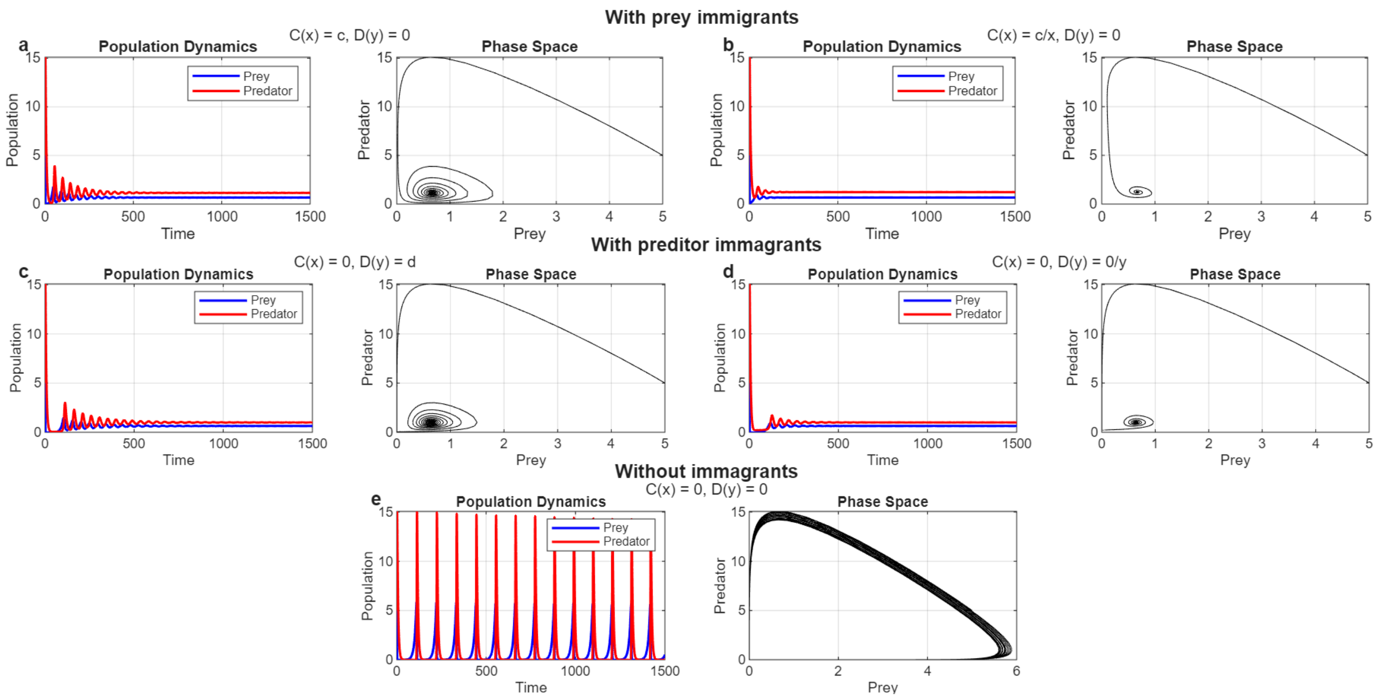

Fig. 1. Recreation of Fig. 1 from [1], generated using an implementation of the linear model described in [1] with identical parameter values: $x_0 = 5$, $y_0 = 5$, $r = 0.1$, $a = 0.1$, $b = 0.3$, $m = 0.2$, $c = 0.01$, $d = 0.01$.

I. Introduction

In Tahara et al. [1], the authors argue that the inherently non-stable dynamics of the classical Lotka–Volterra (LV) system limit its applicability to most real populations. While they reiterate that the classic LV dynamics make it a good match for closed cyclic systems, they assert that there are many observed instances of stable coexistence in wild predator–prey systems that the base model fails to simulate accurately.

Tahara et al. [1] set out to address the non-stable dynamics of Lotka–Volterra systems by adapting the base LV model to represent open systems with stable predator–prey coexistence. They describe the implementation of additional dual-use immigration and migration terms appended to the linear[1] LV model as follows:

$$\begin{cases} \dfrac{dx}{dt} = rx - axy + C(x), \\[2ex] \dfrac{dy}{dt} = bxy - my + D(y), \end{cases} \tag{1}$$

[1]Tahara et al. [1] also briefly explore hyperbolic and sigmoid functional-based LV predator–prey equations; however, both their primary focus and the focus of this paper remain on the linear LV implementation.

These functions, denoted $C(x)$ and $D(y)$ for prey and predators respectively, demonstrated success in stabilizing the system under constant, proportional, and random immigration, while also revealing that migration effects could destabilize the system under certain conditions. This case study seeks to reproduce and explore applications of the linear model presented in [1].

II. Model Reproduction

The first step in this case study was to implement the model described by Tahara et al. [1] in MATLAB. To verify that our implementation behaved as expected, we replicated Fig. 1 from [1]. Figure 1 shows the resulting recreation.

A. Parameters

There are three possible migration terms that can be added to each species in the model:

$$C(x) = \begin{cases} 0, & \text{No prey migration,} \\ c, & \text{Constant prey migration,} \\ \dfrac{c}{x}, & \text{Population-proportional prey migration,} \end{cases} \tag{2}$$

$$D(y) = \begin{cases} 0, & \text{No predator migration,} \\ d, & \text{Constant predator migration,} \\ \dfrac{d}{y}, & \text{Population-proportional predator migration.} \end{cases} \tag{3}$$

The first option represents the baseline case in which no migration occurs. The second option, $C(x) = c$ and $D(y) = d$, represents constant prey and predator immigration, respectively, meaning that a fixed number of individuals is added to a population each cycle. This could model interaction with a neighboring population that continuously exchanges individuals.

The third option, $C(x) = c/x$ and $D(y) = d/y$, represents density-dependent immigration for prey and predators. In this case, migration rates depend on population size, modeling scenarios in which migration intensity increases with population density. Practically, this corresponds to a fixed fraction of the population migrating each cycle.

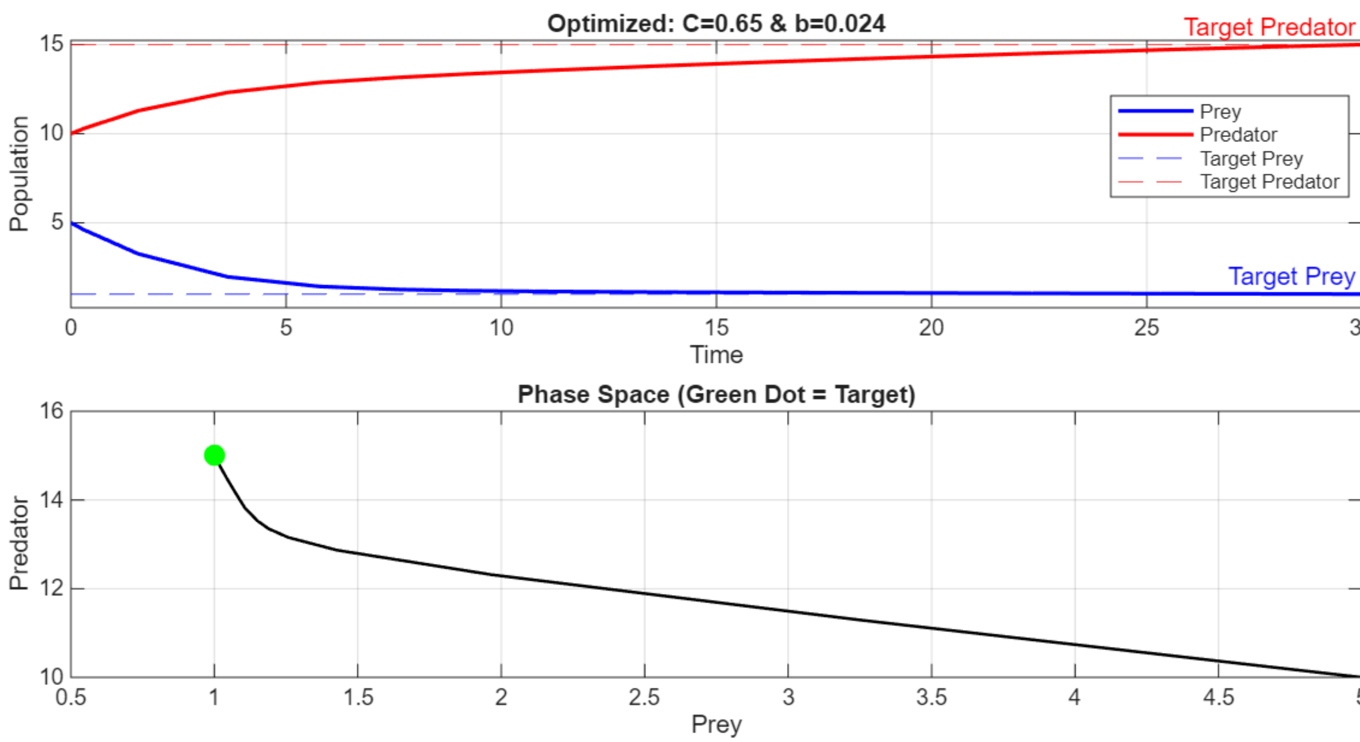

Fig. 2. A Pair of Phase Space and Dynamics Graphs generated from values found using the equilibrium solver algorithm.

III. Arbitrary Equilibrium Tuning

Tasked with developing a method to adjust the equilibrium point of the modified LV system, we designed a function that, given values for prey growth rate, predation rate, predator mortality rate, initial predator and prey populations, and predator immigration multipliers, computes specific values for the predator reproduction rate (b) and constant prey immigration (c, assuming $C(x) = c$). These values ensure that the equilibrium state of the system converges to an arbitrarily specified predator and prey population.

Our function, `solver()`, primarily utilizes MATLAB's `fminsearch()` function. Due to the difficulty of deriving an analytical solution, the optimization process treats the biological simulation as a black box. The solver provides `fminsearch()` with initial guesses for c and b, then iteratively evaluates candidate solutions.

Each iteration passes trial parameters to the `CalculateError()` function, which acts as an interface between the optimizer and the population model. This function computes an error value, which `fminsearch()` uses to determine the next search direction.

Within `CalculateError()`, the `GetDerivatives()` function is executed using an ODE solver (`ode23()`) to simulate population dynamics over a 20-unit time span. Once the simulation concludes, final population values are compared against the target equilibrium, and the sum of squared errors is returned.

The `fminsearch()` algorithm operates via point testing, maintaining a simplex of three points. Based on their associated error values, the simplex is translated, expanded, or contracted until convergence. The final simplex center corresponds to the optimal c and b values. The remainder of the script handles parameter initialization and visualization.

IV. Real-World Applications

One scenario described in the case study instructions involves a forestry service seeking to regulate wildlife populations. In such a case, the service would first observe the ecosystem to estimate current population sizes and reproduction rates. These values would then be provided to the solver, which outputs a value for C (the number of prey to add or remove) and a value for b (the predator reproduction rate), which could be influenced by adjusting available resources.

For example, consider the system

$$\frac{dx}{dt} = rx - axy + C, \qquad \frac{dy}{dt} = bxy - my,$$

with parameters $r = 0.1$, $a = 0.05$, and $m = 0.02$, where each time step represents one day. If the current populations are 5 Prey and 10 Predators (in units of hundreds of thousands), and the desired equilibrium is 1 Prey and 15 Predators. Figure 2 depicts the solver function's output, recommending adding approximately 0.65 predators per week and reducing the predator reproduction rate to 0.02 per week. With these parameters, the LV Model converges on the expected equilibrium quickly and efficiently.

V. Conclusion

Overall, we found success in developing a tuning model for the adjusted LV systems from [1]. While these differential equation models may not be perfectly actuate in the real world, they still certainly provide important insights into how we might understand issues such as prey-predator relationships.

VI. Notes on the Utilization of AI

Artificial intelligence tools, including ChatGPT, were utilized during the planning and development of this case study. These tools assisted with grammar refinement, formatting, reorganization, and logical consistency checks. All generated content was thoroughly reviewed, edited, and verified prior to inclusion. This paper represents the result of substantial intellectual contribution by Ilan Kliman and Rex Paster, and all submitted materials have been carefully reviewed to ensure accuracy and originality.

The following prompts were used during development:

- "Help add labels to the right of each pair of graphs."
- "Help space out each row to make room for titles."
- "Help add titles for each pair of graphs."
- "Help add row titles."
- "How would I put this equation into LaTeX?"
- "Why isn't my BibTeXfile compiling?"
- "Make testing data for this function."
- "Spellcheck."
- "How do I format an IEEE citation?"
- "Can you clean up the formatting and spellcheck the following latex file?"

REFERENCES

[1] T. Tahara *et al.*, "Asymptotic stability of a modified Lotka–Volterra model with small immigrations," *Scientific Reports*, vol. 8, 2018, Art. no. 25436, doi: 10.1038/s41598-018-25436-2.

[2] MathWorks, "How do I use `fminsearch` to find the minimum or maximum of a function $x^4 - 3xy + 2y^2$?," *MATLAB Answers*, accessed Feb. 8, 2026. [Online].

[3] MathWorks, Inc., *MATLAB Documentation*, accessed Feb. 8, 2026. [Online].